LLM ENGINEERING

Fine-tuning LLMs Visual Guide

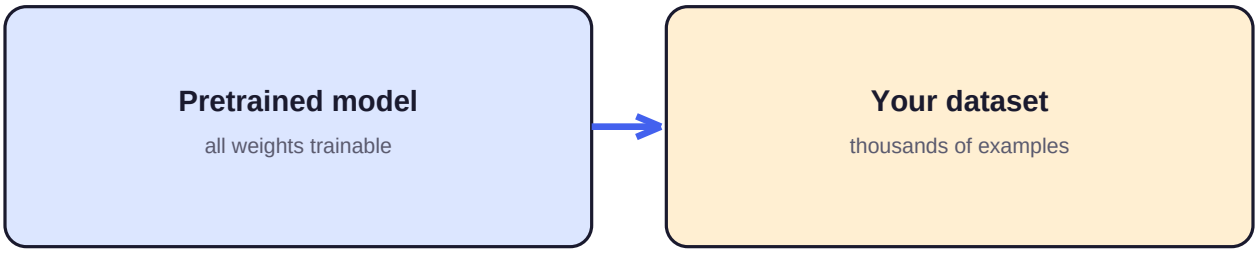
Adapting big models cheaply:
full tuning, LoRA/QLoRA, instruction tuning, RLHF.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Full Fine-tuning

Update every weight. Powerful, expensive.



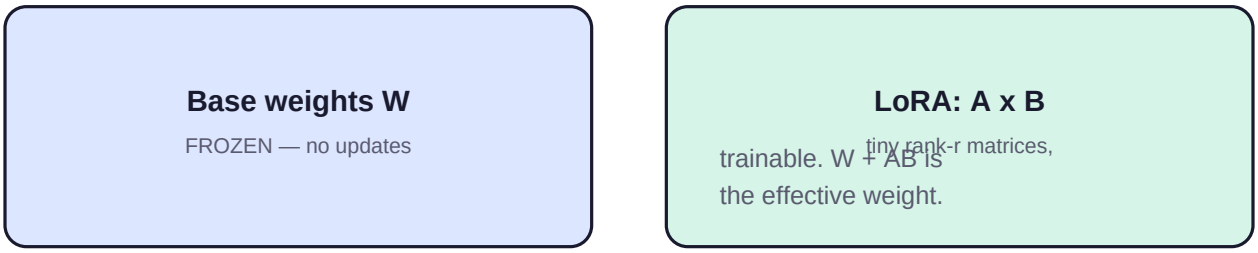
Every parameter updates. Needs serious GPU memory — optimizer states can triple the footprint.

Risks: catastrophic forgetting of general ability, and overfitting on small datasets. Small learning rate, few epochs.

STEP 2/6

LoRA: Low-Rank Adapters

Freeze the giant; train a tiny side-path.



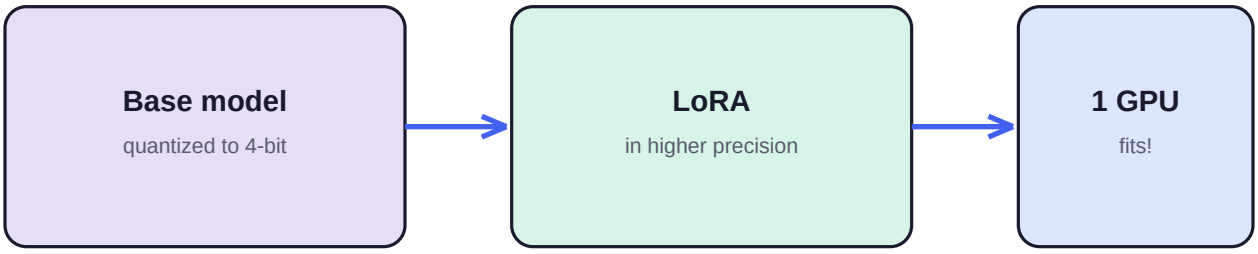
Rank r (8-64) controls capacity. Adapters are <1% of parameters — mergeable at inference for zero overhead.

LoRA adapters are swappable: one base model, many task adapters. The standard efficient fine-tuning method.

STEP 3/6

QLoRA: Quantize the Base

Fine-tune a 65B model on one GPU.



4-bit NormalFloat quantization shrinks the frozen base ~8x. Quality stays close to full fine-tuning on most tasks.

STEP 4/6

Instruction Tuning

Teach the model to follow instructions.

Instruction: 'Summarize this report in 3 bullets'

input

Response: '- Revenue up 12% ...'

target output

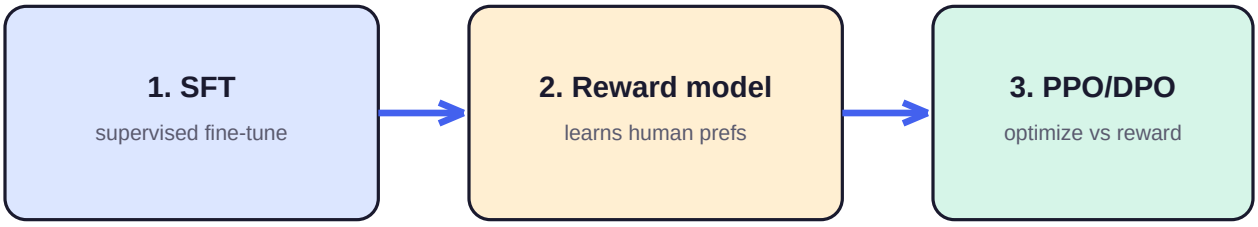
Train on (instruction, response) pairs — FLAN, Alpaca-style datasets. The model learns the follow-instructions behav

Instruction tuning is what turns a raw next-token model into a helpful assistant. Quality and diversity of examples matter most.

STEP 5/6

The RLHF Pipeline

Align with human preferences in three stages.



Humans rank pairs of responses; the reward model generalizes those rankings; PPO/DPO steers the policy toward them.

STEP 6/6

Evaluating After Tuning

Prove the tune helped — and didn't break anything.

- Hold out a test set BEFORE tuning. Report task metrics (accuracy, F1) plus a general benchmark (MMLU-lite).
- Regression-test old capabilities: fine-tunes can silently degrade general knowledge.
- Check for overfitting: huge train/test gaps, or the model parroting training phrasing.
- Human spot-checks on 50-100 outputs catch what metrics miss — tone, format, refusal behavior.

If evals don't move, the bottleneck is data quality — not rank, not epochs. Curate harder examples.

STEP ✓/✓

Key Takeaways

Fine-tuning LLMs — Remember

- Full fine-tuning updates everything: powerful but GPU-hungry and forgetful.
- LoRA trains tiny rank-r adapters on a frozen base — swappable, mergeable, standard.
- QLoRA quantizes the base to 4-bit so huge models fit on one GPU.
- Instruction tuning teaches follow-instructions behavior; RLHF/DPO aligns with human preference.
- Evaluate on held-out tests AND general benchmarks to catch regressions.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev